

## **Overview of Project Management**

**Software Project Management (SPM)** is a proper way of planning and leading software projects. It is a part of project management in which software projects are planned, implemented, monitored, and controlled.

What is Project?

A project is a group of tasks that need to complete to reach a clear result. A project also defines as a set of inputs and outputs which are required to achieve a goal. Projects can vary from simple to difficult and can be operated by one person or a hundred.

Projects usually described and approved by a project manager or team executive. They go beyond their expectations and objects, and it's up to the team to handle logistics and complete the project on time. For good project development, some teams split the project into specific tasks so they can manage responsibility and utilize team strengths.

What is software project management?

Software project management is an art and discipline of planning and supervising software projects. It is a sub-discipline of software project management in which software projects planned, implemented, monitored and controlled.

It is a procedure of managing, allocating and timing resources to develop computer software that fulfills requirements.

In software Project Management, the client and the developers need to know the length, period and cost of the project.

Prerequisite of software project management?

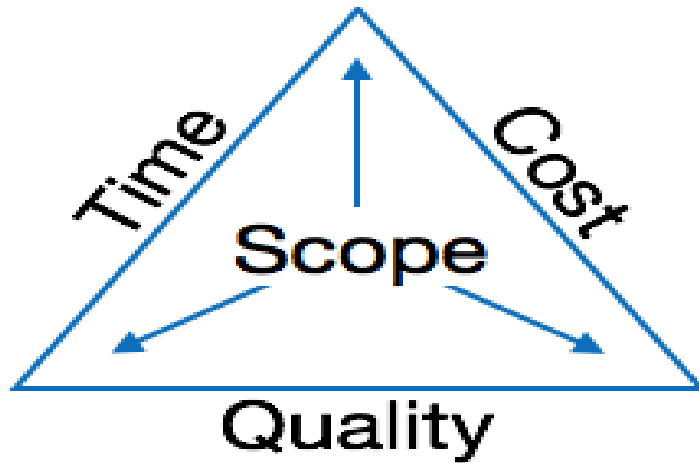
There are three needs for software project management. These are:

1. Time
2. Cost
3. Quality

It is an essential part of the software organization to deliver a quality product, keeping the cost within the client's budget and deliver the project

as per schedule. There are various factors, both external and internal, which may impact this triple factor. Any of three-factor can severely affect the other two.

### **Need of software project management**



The image above shows triple constraints for software projects. It is an essential part of software organization to deliver quality product, keeping the cost within client's budget constrain and deliver the project as per scheduled. There are several factors, both internal and external, which may impact this triple constrain triangle. Any of three factor can severely impact the other two.

Therefore, software project management is essential to incorporate user requirements along with budget and time constraints.

### **Project Manager**

A project manager is a character who has the overall responsibility for the planning, design, execution, monitoring, controlling and closure of a project. A project manager represents an essential role in the achievement of the projects.

A project manager is a character who is responsible for giving decisions, both large and small projects. The project manager is used to manage the risk and minimize uncertainty. Every decision the project manager makes must directly profit their project.

Role of a Project Manager:

### **1. Leader**

A project manager must lead his team and should provide them direction to make them understand what is expected from all of them.

### **2. Medium:**

The Project manager is a medium between his clients and his team. He must coordinate and transfer all the appropriate information from the clients to his team and report to the senior management.

### **3. Mentor:**

He should be there to guide his team at each step and make sure that the team has an attachment. He provides a recommendation to his team and points them in the right direction.

Responsibilities of a Project Manager:

1. Managing risks and issues.
2. Create the project team and assigns tasks to several team members.
3. Activity planning and sequencing.
4. Monitoring and reporting progress.
5. Modifies the project plan to deal with the situation.

## **Factors Influencing project Management**

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In Software Development, task of managing productivity is key issue. The management of software development is heavily dependent on four factors :

1. **People** – When we refer to people in software development, we actually mean hierarchy of roles in software development such as managers, team leaders, senior developers etc. They are ones who form development team and work in development of software. The order of priority of people among other dependency factors is **first** (highest). Software development is people-centric activity. Software development requires good managers, people who understand psychology of people and provide good leadership. A good manager does not ensure success

of project, but can increase probability of success. A good manager gives priority to areas like proper selection, training, compensation, career development, work culture, development team's efficiency etc. Therefore, Managers are crucial and critical for software development and it is their responsibility to manage, motivate, encourage, control and guide his/her development team through tough situations. Hence, success of project is on shoulders of people who are involved in development.

2. **Product** – A product is solution of problem that is handed to user who is struggling with problem. The priority of product among other dependency factors is **second** that is after **people**. The product or solution to problem is constructed based on requirements, which are objectives and scope work of problem. This helps managers to select best approach within constraints imposed by delivery deadlines, budgetary restrictions, personnel availability, technical interfaces etc. Without well defined requirements, it may be impossible to define reasonable estimates of cost, development time and schedule for project.
3. **Process** – The process is way in which we produce software. It provides framework from which comprehensive plan for software development can be established. If process is weak, end product will undoubtedly suffer. The priority of process among other dependency factors is **third**, after **people** and **product**, however, it plays very critical role for success of project. There are many life cycle models and process improvements models. A suitable process model is determined on basis of requirements specification, development team, user, type of project and associated risks of project. A number of different task sets, milestones, work products, and quality assurance points, enable process framework activities to be adopted to characteristics of project and requirements of project team.
4. **Project** – A proper planning is required to monitor status of development and to control complexity. For project to be successful, it should be within budget, follow policies and requirements defined, and completed within timeline stated. Most of projects overrun cost by 100% of budget and are not delivered on time. The priority of project among other dependency factors is **last**(lowest). In order to manage successful project, we must understand and prevent associated risks. Concrete requirements should be defined and frozen, wherever possible. Unnecessary changes should not be made to avoid software surprises. Software surprises are always risky and should be minimized. A planning mechanism should be constructed to caution from occurrence of any surprises.

**All four factors: People, Product, Process and Project are important for success of project. Their relative importance helps us organize development activities in more scientific and professional way.**

## Project Management Activities

Software Project Management consists of many activities, that includes planning of the project, deciding the scope of product, estimation of cost in different terms, scheduling of tasks, etc.

**The list of activities are as follows:**

1. Project planning and Tracking
2. Project Resource Management
3. Scope Management
4. Estimation Management
5. Project Risk Management
6. Scheduling Management
7. Project Communication Management
8. Configuration Management

Now we will discuss all these activities -

**1. Project Planning:** It is a set of multiple processes, or we can say that it a task that performed before the construction of the product starts.

**2. Scope Management:** It describes the scope of the project. Scope management is important because it clearly defines what would do and what would not. Scope Management create the project to contain restricted and quantitative tasks, which may merely be documented and successively avoids price and time overrun.

**3. Estimation management:** This is not only about cost estimation because whenever we start to develop software, but we also figure out their size(line of code), efforts, time as well as cost.

If we talk about the size, then Line of code depends upon user or software requirement.

If we talk about effort, we should know about the size of the software, because based on the size we can quickly estimate how big team required to produce the software.

If we talk about time, when size and efforts are estimated, the time required to develop the software can easily determine.

And if we talk about cost, it includes all the elements such as:

- Size of software
- Quality
- Hardware
- Communication
- Training
- Additional Software and tools
- Skilled manpower

**4. Scheduling Management:** Scheduling Management in software refers to all the activities to complete in the specified order and within time slotted to each activity. Project managers define multiple tasks and arrange them keeping various factors in mind.

**For scheduling, it is compulsory -**

- Find out multiple tasks and correlate them.
- Divide time into units.
- Assign the respective number of work-units for every job.
- Calculate the total time from start to finish.
- Break down the project into modules.

**5. Project Resource Management:** In software Development, all the elements are referred to as resources for the project. It can be a human resource, productive tools, and libraries.

**cResource management includes:**

- Create a project team and assign responsibilities to every team member
- Developing a resource plan is derived from the project plan.
- Adjustment of resources.

**6. Project Risk Management:** Risk management consists of all the activities like identification, analyzing and preparing the plan for predictable and unpredictable risk in the project.

Several points show the risks in the project:

- The Experienced team leaves the project, and the new team joins it.
- Changes in requirement.
- Change in technologies and the environment.
- Market competition.

**7. Project Communication Management:** Communication is an essential factor in the success of the project. It is a bridge between client, organization, team members and as well as other stakeholders of the project such as hardware suppliers.

From the planning to closure, communication plays a vital role. In all the phases, communication must be clear and understood. Miscommunication can create a big blunder in the project.

**8. Project Configuration Management:** Configuration management is about to control the changes in software like requirements, design, and development of the product.

The Primary goal is to increase productivity with fewer errors.

**Some reasons show the need for configuration management:**

- Several people work on software that is continually update.
- Help to build coordination among suppliers.
- Changes in requirement, budget, schedule need to accommodate.
- Software should run on multiple systems.

**Tasks perform in Configuration management:**

- Identification
- Baseline
- Change Control
- Configuration Status Accounting
- Configuration Audits and Reviews

## People involved in Configuration Management:



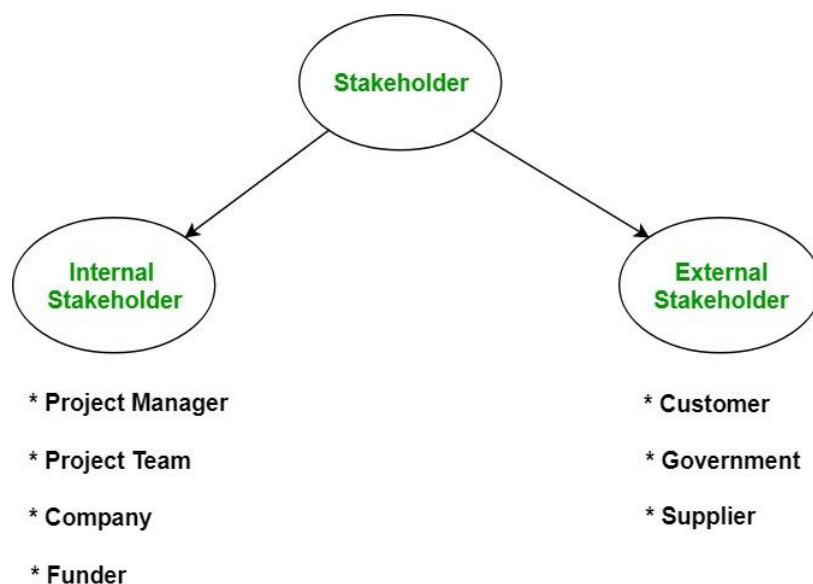
## Stakeholder

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In simple words, anyone having any type of relation/interest in the project is known as stakeholder. The term **Software Project Stakeholder** refers to, "a person, group or company that is directly or indirectly involved in the project and who may affect or get affected by the outcome of the project".

### What is Stakeholder Identification?

It is the process of identifying a person, group or a company which can affect or get affected by a decision, activity or the outcome of the software project. It is important in order to identify the exact requirements of the project and what various stakeholders are expecting from the project outcome.



Type of Stakeholders:

**1. Internal Stakeholder:**

An internal stakeholder is a person, group or a company that is directly involved in the project.

For example,

**1. Project Manager:**

Responsible for managing the whole project. Project Manager is generally never involved in producing the end product but he/she controls, monitors and manages the activities involved in the production.

**2. Project Team:**

Performs the actual work of the project under the Project Manager including development, testing, etc.

**3. Company:**

Organisation who has taken up the project and whose employees are directly involved in the development of the project.

**4. Funders:**

Provides funds and resources for the successful completion of the project.

**2. External Stakeholder:**

An external stakeholder is the one who is linked indirectly to the project but has significant contribution in the successful completion of the project.

For example,

**1. Customer:**

Specifies the requirements of the project and helps in the elicitation process of the requirement gathering phase. Customer is the one for whom the project is being developed.

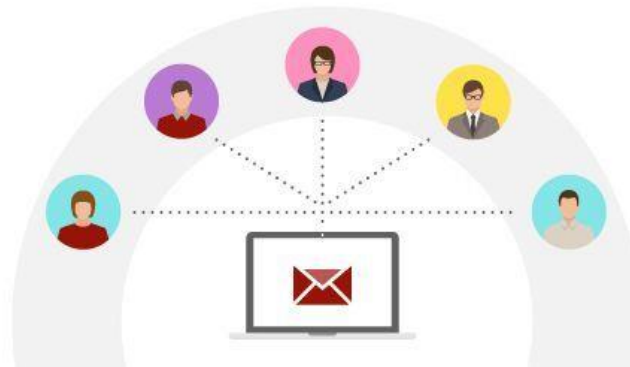
**2. Supplier:**

Supplies essential services and equipment for the project.

**3. Government:**

Makes policies which helps in better working of the organisation.

## Communication Management



Communication management in a project situation is the most important aspect, which aims at ensuring that “right information reaches the right stakeholder at the right time in the right manner” so that effective and efficient exchange of information takes place, resulting in greater collaboration between all stakeholders. Stakeholder collaboration and engagement is critical for project success.

As per PMBOK guide, “Communication management includes the processes required to ensure timely and appropriate generation, collection, distribution, storage, retrieval and ultimate disposition of project information.” Communication is a two way process of sending information from one entity to another. The Project Manager spends more than 90% of time in Project Communication.

For managing communication, the project manager and the project team members need to possess excellent communication and interpersonal skills. These skills will ensure the both sides in a communication understand each other clearly. Some of the communication skills and interpersonal skills needed include the following:

- Written communication ability
- Verbal communication ability
- Non-verbal communication skills
- Listening skills
- Empathy for the stakeholders
- Questioning and probing skills

- Influencing ability
- Interpersonal skills (Conflict management)
- Political and social skills

In addition to having the above skills, we also need to adopt a structured approach for managing communication. Project communication management includes the following 3 processes as per PMBOK guide, 6th edition.



- **Plan Communication Management** – It is first important to develop a communication management plan or approach. This includes understanding the information requirement for each stakeholder or stakeholder group. Also understanding each stakeholder or stakeholder groups' communication preferences such as language, medium, technology, frequency, format if any for each information exchange. During communication planning, the team also develops a strategy for information exchange using different communication methods such as “interactive, push and pull” methods. Understanding the social and cultural background of each stakeholder is also done and accordingly communication approach is prepared.
- **Manage Communication** – Communication starts happening throughout the project by following the communication management plan. The required information is collected, created, distributed, and stored for future reference throughout the project life cycle fulfilling the information needs of all stakeholders. Here the project manager and the team members responsible for communication use their communication skills.

- **Monitor Communication** – While communication starts in the project, it is also important to keep ensuring that the information needs of all stakeholders are met. Any issues of stakeholders pertaining to communication and their information needs are addressed by adjusting the communication plan. This is also an on-going parallel process which runs throughout the project life cycle.

## **The 5 Phases of Project Management**

- 1.Conception & Initiation
- 2.Definition & Planning
- 3.Launch or Execution
- 4.Performance & Control
- 5.Project Closure

### **Phase 1: Conception & Initiation**

- Build a strong case for it (Why are we doing this and what are the benefits?)
- Feasibility study/analysis (Can we do it? How much time and resources will it require?)
- Identify key stakeholders.
- Develop a project charter. A project charter is a document that offers a formal initiation definition of the project and authorized the project manager to use the company's organizational resources for project activities.

### **Phase 2: Project Definition & Project Planning**

A project management plan defines how a project will be executed, monitored, controlled and closed basically, covering all the consequent stages of a project. Remember, your plan should also account for the likely changes in the project environment e.g. when new information becomes available, or if any requirements change.

- **Project scope management plan** documenting all the deliverables of the project; scope definition, validation and control. It should present a

detailed list of requirements for the project and a bird-eye view of what should be delivered and when.

- **Requirements management plan** indicates how the requirements activities will be planned, tracked and reported. It includes all the respective project management process steps you will have to take to initiate changes, prioritize different requirements, track and report them.
- Some managers choose to visualize this part using various project management processes charts and graphs e.g. like this one:
- **Constraints:** information about the known bottlenecks e.g. limited human resources, short timeline. Some managers also choose to include a risk management plan a document detailing individual project risks; assessing their probability of occurrence and the possible effect they may cast on different project processes; and outlines how those would be mitigated.
- **Human resources requirements:** the assigned project team, roles and responsibilities of each member.
- **Project schedule and key milestones:** this section should outline the key project deliverables, due date or duration of each phase/iteration and key dependencies between them. The easiest way to illustrate this point is to use a Gantt chart.
- **Budget/cost estimates:** a preliminary assessment of the funds required for the project. The costs are typically divided into three categories: capital items, expense items and labor

### **Phase 3: Project Execution**

.Assemble a team

- Assign resources
- Start executing the project management plan
- Proactive control and manage project execution
- Identify KPIs and set-up tracking systems.
- Distribute and assign tasks.

- Schedule status meetings
- Update project schedule
- Modify project plans as needed

#### **Phase 4: Project Performance & Project Controlling**

- Cost tracking: review and refine project costs to reflect additional details as those become available.
- Quality management: make sure that organizational quality policies are incorporated into the project. All the deliverables are produced according to the predefined specs.
- Overall project performance monitoring: monitor and respond to changes in the project. Assess the team effectiveness, proactively identify and prevent project bottlenecks and ensure that all the project 'cogs' are moving without any friction. This also includes sharing project status reports with key stakeholders upon reached milestones.

#### **Phase 5: Project Closure**

professional PMs will often schedule a closing meeting the so-called "project post-mortem" to draw the final line. Such meetings are held to discuss the overall project performance, identify key failures and successes and apply those insights towards future projects.

#### **What is a project charter?**

A project charter is a formal short document that states a project exists and provides project managers with written authority to begin work. A project charter document describes a project to create a shared understanding of its goals, objectives and resource requirements before the project is scoped out in detail.

Project charters are an important part of project management, as they help plan out the basics of a project and can be referenced throughout the project's lifecycle. The formal document can also show a project's viability and possible return on investment, helping the work get approved.

The project charter helps project managers explain to participants and stakeholders the scope of a project, project objectives, who will participate in the project, along with other details such as possible risks. Depending on a company's culture and management style, a charter may serve the same purpose as a business case. In a large organization, the charter may be a multipage document, but in smaller organizations, it may just be a few paragraphs with bulleted items.

### **What should a project charter include?**

Project charter templates often include the following parts of the project:

- **Project goal.** This documents the reasons for undertaking the project in clear, concise language. This should determine the project's scope.
- **Project participants.** This identifies who will be involved in the project and clearly states their roles.
- **Stakeholders.** This identifies project sponsors or other people who will be directly affected by the project and need to know about its progress.
- **Requirements.** This identifies the resources required for the project's objectives to be achieved.
- **Constraints.** This documents potential roadblocks or bottlenecks and should help prepare participants for the potential issues of the project.
- **Implementation milestones.** This identifies the start and ideal completion dates, as well as dates for other potentially important checkpoints, like a project schedule.
- **Communication.** This specifies how the project manager will communicate with project owners, participants and key stakeholders throughout the project.
- **Deliverables.** This documents what specific products, processes or services the project provides upon completion.
- **Cost.** This identifies a general overview of the project budget.

The project manager should create the project charter at the beginning of the project. Signing the document can also act as a way to authorize the project -- giving the project manager the go-ahead to officially begin executing the project. This includes the authority to begin employing organizational funds and resources. For larger projects that are multi phased , project managers can create a charter for each phase.

### **What is Statement of Work (SOW)?**

A statement of work (SOW) is a document that provides a description of a given project's requirements. It defines the scope of work being provided, project deliverables, timelines, work location, and payment terms and conditions.

Statement of work management is simply the process for ensuring the agreed-upon scope of services within the SOW are being completed on time and on budget.

Whomever is responsible for SOW management is responsible for creating efficiencies, risk mitigation, any special requirements, and supplier management and negotiations. Ultimately, finding savings opportunities and reporting on a project's overall success.

SOWs are commonly used along with a:

### **Request for Proposal (RFP)**

Organisations typically use this document to request pricing, provide requirements, and define a period of performance. Enough information that a service provider can respond back with an accurate bid for a project.

### **Master Services Agreement (MSA)**

This is the governing document that outlines two parties' terms and conditions for an entire relationship. The statement of work document usually only covers details belonging to a single project or scope of work.

### **How are SOWs created?**

There are many SOW templates available to help get managers started. However, to write a statement of work can be somewhat complex. Getting everything included to ensure the scope of services is detailed enough is critical.

Elements of an SOW can include:

- Purpose of the project

- Scope of work being performed
- Location of the project, project length, and any work requirements
- Expected deadlines and deliverables
- Acceptance criteria
- Any hardware and software required
- Performance-based standards to be met

Creating a thorough SOW can eliminate the risk that can come if there are any misunderstandings or disputes between parties on any of the elements above.

A concise and well written SOW, mitigates the risk of overspend by ensuring both supplier and organisation have a clear understanding of, and accountability for, the work involved. Bolstering the upfront agreement, lessens the opportunity for misunderstandings resulting in contract extensions and associated costs.

## **What are the different types of SOWs?**

There are three standard SOW categories that companies use for defining scope and procuring services. Some are more popular than others within certain industries, but they all have similar components that define its success.

### **1. Design or detail statement of work**

This category of SOW defines the exact requirements needed to complete a project. It tells the supplier exactly **how to do the work** and what processes to follow.

Additionally, it defines all requirements and any specific industry-related regulations that must be followed by the contractors.

Typically, the organisation using design SOWs assumes most of the risk for the project.

### **2. Level of effort**

This SOW is used for any kind of service. In a very general way, it details work hours and any material needed to perform the service over a given time.

### **3. Performance-based statement of work**

A performance-based SOW clearly lays out the project's purpose, resources that will be provided, and deliverables that will be accomplished. But **it does not** provide details about how the work needs to be performed.

This is the preferred SOW for most companies. This SOW offers the

most flexibility, focuses on project outcomes, and shares risk between parties.

### **What are the benefits of managing SOWs?**

- Benefits can include:
- Increased cost savings opportunities
- Supplier performance and risk mitigation
- Greater process efficiencies
- Detailed reporting
- Project performance management
- Visibility into all outsourced projects within a single purview
- Improved workforce management
- Organisational compliance and risk mitigation

### **Project Planning**

Before starting a software project, it is essential to determine the tasks to be performed and properly manage allocation of tasks among individuals involved in the software development. Hence, planning is important as it results in effective software development.

Project planning is an organized and integrated management process, which focuses on activities required for successful completion of the project. It prevents obstacles that arise in the project such as changes in projects or organization's objectives, non-availability of resources, and so on. Project planning also helps in better utilization of resources and optimal usage of the allotted time for a project. The other objectives of project planning are listed below.

- It defines the roles and responsibilities of the project management team members.
- It ensures that the project management team works according to the business objectives.
- It checks feasibility of the schedule and user requirements.
- It determines project constraints.

Several individuals help in planning the project. These include senior management and project management team. Senior management is responsible for employing team members and providing resources required for the project. The project management team, which generally includes project managers and developers, is responsible for planning, determining, and tracking the activities of the project. Table lists the tasks performed by individuals involved in the software project.

## Tasks of Individuals involved in Software Project

| Senior Management                                                                                                                                                                                                                                                                                                                                                                                     | Project Management Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Approves the project, employ personnel, and provides resources required for the project.</li><li>• Reviews project plan to ensure that it accomplishes the business objectives.</li><li>• Resolves conflicts among the team members.</li><li>• Considers risks that may affect the project so that appropriate measures can be taken to avoid them.</li></ul> | <ul style="list-style-type: none"><li>• Reviews the project plan and implements procedures for completing the project.</li><li>• Manages all project activities.</li><li>• Prepares budget and resource allocation plans.</li><li>• Helps in resource distribution, project management, issue resolution, and so on.</li><li>• Understands project objectives and finds ways to accomplish the objectives.</li><li>• Devotes appropriate time and effort to achieve the expected results.</li><li>• Selects methods and tools for the project.</li></ul> |

Project planning should be effective so that the project begins with well-defined tasks. Effective project planning helps to minimize the additional costs incurred on the project while it is in progress. For effective project planning, some principles are followed. These principles are listed below.

- **Planning is necessary:** Planning should be done before a project begins. For effective planning, objectives and schedules should be clear and understandable.
- **Risk analysis:** Before starting the project, senior management and the project management team should consider the risks that may affect the project. For example, the user may desire changes in requirements while the project is in progress. In such a case, the estimation of time and cost should be done according to those requirements (new requirements).
- **Tracking of project plan:** Once the project plan is prepared, it should be tracked and modified accordingly.
- **Meet quality standards and produce quality deliverables:** The project plan should identify processes by which the project management team can ensure quality in software. Based on the process selected for ensuring quality, the time and cost for the project is estimated.

- **Description of flexibility to accommodate changes:** The result of project planning is recorded in the form of a project plan, which should allow new changes to be accommodated when the project is in progress.

Project planning comprises project purpose, project scope, project planning process, and project plan. This information is essential for effective project planning and to assist project management team in accomplishing user requirements.

## **Work Breakdown Structure(WBS)**

Dividing complex projects to simpler and manageable tasks is the process identified as Work Breakdown Structure (WBS). Usually, the project managers use this method for simplifying the project execution. In WBS, much larger tasks are broken down to manageable chunks of work. These chunks can be easily supervised and estimated. WBS is not restricted to a specific field when it comes to application. This methodology can be used for any type of project management.

Following are a few reasons for creating a WBS in a project:

- Accurate and readable project organization.
- Accurate assignment of responsibilities to the project team.
- Indicates the project milestones and control points.
- Helps to estimate the cost, time and risk.
- Illustrate the project scope, so the stakeholders can have a better understanding of the same.

## **Construction of a WBS**

Identifying the main deliverables of a project is the starting point for deriving a work breakdown structure.

This important step is usually done by the project managers and the subject matter experts (SMEs) involved in the project. Once this step is completed, the subject matter experts start breaking down the high-level tasks into smaller chunks of work.

In the process of breaking down the tasks, one can break them down into different levels of detail. One can detail a high-level task into ten sub-tasks while another can detail the same high-level task into 20 sub-tasks.

Therefore, there is no hard and fast rule on how you should breakdown a task in WBS. Rather, the level of breakdown is a matter of the project type and the management style followed for the project.

In general, there are a few "rules" used for determining the smallest task chunk. In "two weeks" rule, nothing is broken down smaller than two weeks worth of work.

This means, the smallest task of the WBS is at least two-week long. 8/80 is another rule used when creating a WBS. This rule implies that no task should be smaller than 8 hours of work and should not be larger than 80 hours of work.

One can use many forms to display their WBS. Some use tree structure to illustrate the WBS, while others use lists and tables. Outlining is one of the easiest ways of representing a WBS.

Following example is an outlined WBS:

|              |        |             |                    |
|--------------|--------|-------------|--------------------|
| Project Name | Task 1 | Subtask 1.1 | Work Package 1.1.1 |
|              |        |             | Work Package 1.1.2 |
|              |        | Subtask 1.2 | Workpackage 1.2.1  |
|              |        |             | Workpackage 1.2.2  |
|              | Task 2 | Subtask 2.1 |                    |
|              |        |             | Workpackage 2.1.1  |
|              |        |             | Workpackage 2.1.2  |
|              |        |             |                    |

There are many design goals for WBS. Some important goals are as follows:

- Giving visibility to important work efforts.
- Giving visibility to risky work efforts.
- Illustrate the correlation between the activities and deliverables.
- Show clear ownership by task leaders.

## WBS Diagram

In a WBS diagram, the project scope is graphically expressed. Usually the diagram starts with a graphic object or a box at the top, which represents the entire project. Then, there are sub-components under the box.

These boxes represent the deliverables of the project. Under each deliverable, there are sub-elements listed. These sub-elements are the activities that should be performed in order to achieve the deliverables.

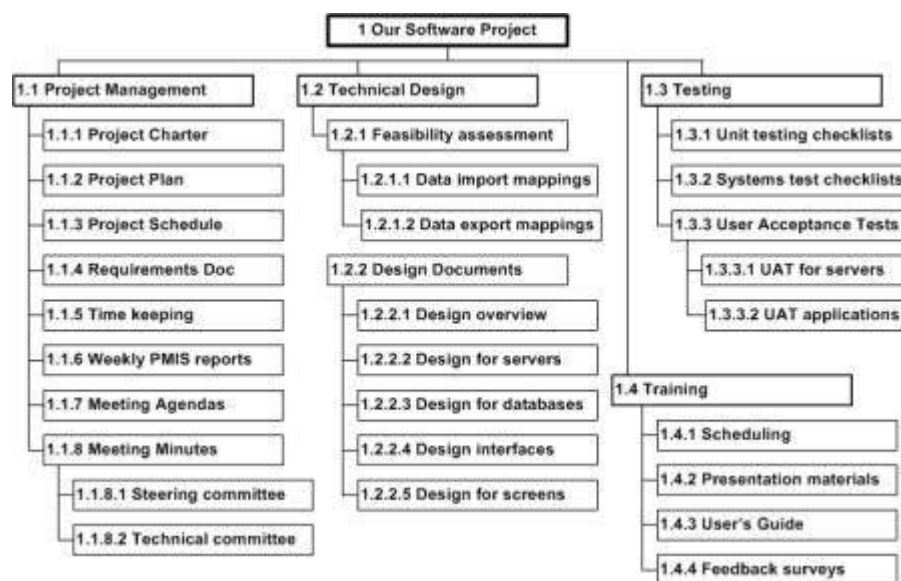
Although most of the WBS diagrams are designed based on the deliveries, some WBS are created based on the project phases. Usually, information technology projects are perfectly fit into WBS model.

Therefore, almost all information technology projects make use of WBS.

In addition to the general use of WBS, there is specific objective for deriving a WBS as well. WBS is the input for Gantt charts, a tool that is used for project management purpose.

Gantt chart is used for tracking the progression of the tasks derived by WBS.

Following is a sample WBS diagram:



## Conclusion

The efficiency of a work breakdown structure can determine the success of a project.

The WBS provides the foundation for all project management work, including, planning, cost and effort estimation, resource allocation, and scheduling.

Therefore, one should take creating WBS as a critical step in the process of project management.

## SDLC Cycle

SDLC Cycle represents the process of developing software. SDLC framework includes the following steps:



The stages of SDLC are as follows:

### Stage1: Planning and requirement analysis

Requirement Analysis is the most important and necessary stage in SDLC. The senior members of the team perform it with inputs from all the stakeholders and domain experts or SMEs in the industry. Planning for the quality assurance requirements and identifications of the risks associated with the projects is also done at this stage.

Business analyst and Project organizer set up a meeting with the client to gather all the data like what the customer wants to build, who will be the end user, what is the objective of the product. Before creating a product, a core understanding or knowledge of the product is very necessary.

**For Example**, A client wants to have an application which concerns money transactions. In this method, the requirement has to be precise like what kind of operations will be done, how it will be done, in which currency it will be done, etc.

Once the required function is done, an analysis is complete with auditing the feasibility of the growth of a product. In case of any ambiguity, a signal is set up for further discussion.

Once the requirement is understood, the SRS (Software Requirement Specification) document is created. The developers should thoroughly follow this document and also should be reviewed by the customer for future reference.

### **Stage2: Defining Requirements**

Once the requirement analysis is done, the next stage is to certainly represent and document the software requirements and get them accepted from the project stakeholders.

This is accomplished through "SRS"- Software Requirement Specification document which contains all the product requirements to be constructed and developed during the project life cycle.

### **Stage3: Designing the Software**

The next phase is about to bring down all the knowledge of requirements, analysis, and design of the software project. This phase is the product of the last two, like inputs from the customer and requirement gathering.

### **Stage4: Developing the project**

In this phase of SDLC, the actual development begins, and the programming is built. The implementation of design begins concerning writing code. Developers have to follow the coding guidelines described by their management and programming tools like compilers, interpreters, debuggers, etc. are used to develop and implement the code.

### **Stage5: Testing**

After the code is generated, it is tested against the requirements to make sure that the products are solving the needs addressed and gathered during the requirements stage. During this stage, unit testing, integration testing, system testing, acceptance testing are done.

### **Stage6: Deployment**

Once the software is certified, and no bugs or errors are stated, then it is deployed. Then based on the assessment, the software may be released as it

is or with suggested enhancement in the object segment. After the software is deployed, then its maintenance begins.

## **Stage7: Maintenance**

Once when the client starts using the developed systems, then the real issues come up and requirements to be solved from time to time. This procedure where the care is taken for the developed product is known as maintenance.

### **There are five generic process framework activities:**

#### **1. Communication:**

The software development starts with the communication between customer and developer.

#### **2. Planning:**

It consists of complete estimation, scheduling for project development and tracking.

#### **3. Modeling:**

- Modeling consists of complete requirement analysis and the design of the project like algorithm, flowchart etc.
- The algorithm is the step-by-step solution of the problem and the flow chart shows a complete flow diagram of a program.

#### **4. Construction:**

- Construction consists of code generation and the testing part.
- Coding part implements the design details using an appropriate programming language.
- Testing is to check whether the flow of coding is correct or not.
- Testing also check that the program provides desired output.

#### **5. Deployment:**

- Deployment step consists of delivering the product to the customer and take feedback from them.
- If the customer wants some corrections or demands for the additional capabilities, then the change is required for improvement in the quality of the software.

